

Root Cause Analysis Report

INC-003 — MIF Interface Failure: Work Order Cost Data Not Reaching Finance System

Incident ID	INC-003
System	IBM Maximo (CAFM/EAM) — Work Order & Integration modules
Severity	Critical — financial reconciliation impact
Detected	During month-end reconciliation
Prepared by	CAFM Application Support Engineer (simulated work sample)

1. Summary

Completed Work Order cost data stopped flowing from Maximo to the downstream finance system without any visible error to end users. The issue was only surfaced during month-end reconciliation, when finance identified a gap between Maximo-reported completed work order costs and what had landed in the finance system.

2. Impact

Approximately 11 days of completed work order cost transactions (est. 340 transactions) did not reach the finance system's ledger, delaying month-end close reconciliation and requiring manual cost verification for the affected period.

3. Timeline

Day 0: External finance system deploys a schema change to the cost center code field (numeric → alphanumeric), without prior notice to the integration/CAFM team.

Day 0 onward: Outbound MIF interface (WOCOST_OUT) begins failing validation silently for every completed work order with a cost center reference; transactions queue as ERROR status rather than raising an active alert.

Day 11: Finance flags a reconciliation gap during month-end close.

Day 11 (+2 hrs): Support engineer identifies and confirms root cause.

Day 11 (+5 hrs): Fix applied, failed transactions reprocessed, finance confirms reconciliation restored.

4. Root Cause

The MIF outbound processing rule for the WOCOST_OUT interface validated the cost center field against a numeric format. The external finance system changed this field to alphanumeric without notifying the integration owners. Every transaction with a cost center reference after the change failed MIF field-level validation and was written to the interface error table with status ERROR — a state with no active alerting attached, so it was not noticed until the monthly reconciliation cycle surfaced the gap.

5. Diagnostic Evidence

Interface errors were confirmed directly from the MIF error table:

```
SELECT intfacename, status, transdate, extsysname, message
FROM maxintfile
WHERE status = 'ERROR' AND intfacename = 'WOCOST_OUT'
```

```
ORDER BY transdate DESC;
```

This returned a consistent field-validation error message referencing the cost center field, with a first-occurrence timestamp matching the finance system's deployment window.

6. Corrective Action

1. Updated the MIF outbound processing rule's field validation to accept the new alphanumeric cost center format.
2. Reprocessed all queued ERROR-status transactions from the affected window.
3. Verified with finance that reprocessed totals reconciled correctly against expected work order cost figures.

7. Preventive Measures

1. Built an interface health check querying MAXINTFILE status counts by interface name, with alerting triggered on any ERROR count above zero, rather than relying on downstream teams to notice a gap during reconciliation.
2. Raised a change-management requirement with the finance system's team: any field-level schema change on either side of an active integration must be communicated to the integration owner before deployment.
3. Added WOCOST_OUT to the list of interfaces reviewed during monthly integration health audits.